SatQuery AI — Feature Verification Guide

Comprehensive Test Suites, Multimodal Input Imagery, Prompts, and Expected Execution Results

Project: NOVEX / SatQuery AI • Architecture: Multimodal Remote-Sensing Agent

Target Platform	SatQuery AI Full-Stack (Next.js 14 + FastAPI + GIS Engine)
Live Vercel URL	https://novex-teal.vercel.app/
Local Data Directory	data/demo/ (Georeferenced GeoTIFFs, EPSG:32643)
Supported Modalities	Optical RGB, 12-Band Multispectral, Sentinel-1 C-SAR (Radar), Bi-temporal Pairs

1. Test Case: Single-Image Visual Question Answering (VQA)

Validates automatic modality extraction (RGB Optical, 3 bands, 10m spatial resolution) and scene interpretation without visual hallucination.

Input Image	`data/demo/single/optical/sample_optical.tif`
Sample Prompt	*"What is visible in this satellite image?"*
Model Selected	SatQuery Vision Agent (InternVL3 / Qwen-Vision)
Expected Output	Detailed land cover breakdown: identifies mixed vegetation, transit corridors, and rural-urban interface.
Confidence Score	**0.91 (High)** • Input Quality: Good (0% cloud shadow interference)

2. Test Case: Visual Grounding (Spatial Feature Localization)

Tests the model's ability to locate distinct geographic features and output precise pixel and WGS84 bounding boxes.

Input Image	`data/demo/single/optical/sample_optical.tif`
Sample Prompt	*"Where is the water body located in this area?"*
Output Evidence	Target Bounding Box + GeoJSON Polygon overlaid on satellite map view.
Geographic Bounds	**18.5350° N, 73.8700° E to 18.5600° N, 73.8950° E** (WGS84 reprojected)
Calculated Area	Surface water footprint: **42,000 m² (4.2 hectares)**

3. Test Case: Bi-Temporal Change Detection & Area Quantification

Tests the Siamese change detection engine (ChangeFormer) with two co-registered time-series satellite passes. Verifies real GIS metric calculations (never fabricated percentages).

Input Images (2)	`1. data/demo/temporal/before/sample_before.tif (Date T1)` `2. data/demo/temporal/after/sample_after.tif (Date T2)`
Sample Prompt	*"Did the built-up area increase between these two dates?"*

Task Pipeline	Task Router ➔ Raster Alignment ➔ ChangeFormer Siamese Network ➔ GIS Area Integrator
Quantitative Answer	**+14.8 hectares (+12.3% of AOI)** increase in impervious commercial/residential built-up area.
Generated Evidence	Before/After dual-view composite + Binary Change Mask highlight (Red/Yellow)

4. Test Case: Optical + SAR (Radar) Multimodal Sensor Fusion

Fuses visual spectrum imagery (Sentinel-2) with cloud-penetrating synthetic aperture radar (Sentinel-1 C-band SAR) to extract structures invisible to optical sensors alone.

Input Images (2)	`1. data/demo/fusion/optical/sample_fusion_optical.tif (RGB Optical)` `2. data/demo/fusion/sar/sample_fusion_sar.tif (C-band SAR GRD)`
Sample Prompt	*"Use optical and SAR together to identify built-up and water boundaries."*
Model Adapter	**CROMA (Cross-Modal Multitask Architecture)**
Modality Agreement	**agree** — Radar double-bounce backscatter aligns with optical roof reflectance within 0.15 px.
Key Capability	Separates smooth specular water (-22.4 dB backscatter) from cloud shadow ambiguity.

5. Test Case: Multilingual Query Understanding (Hindi & Hinglish)

Evaluates native multilingual intent parsing using Qwen3 tokenizer across English, Hindi, and Hinglish vernacular phrases.

Hindi Prompt	*"■■ ■■■■■■■ ■■■ ■■■ ■■■■■ ■■ ■■■■■■■■ ■■■■■ ■■?"*
Detected Language	hi (Hindi) ➔ Routes to Spatial Grounding task
Hinglish Prompt	*"Is image me water body aur construction kahan hai?"*
Detected Language	hinglish (Mixed Vernacular) ➔ Auto-normalizes to dual-intent extraction

6. Test Case: Policy Guardrails & Safety Enforcement (Negative Tests)

Verifies that the policy validator strictly rejects incompatible tasks and never hallucinates uncomputed metrics.

Single Image Change	Upload **only 1 image** and prompt: *"What changed between these dates?"* **Expected:** System refuses execution with requires_user_input explaining that change detection strictly requires a before and after pair.
Modality Mismatch	Upload **2 optical images** and prompt: *"Use optical and SAR together."* **Expected:** System identifies both images as optical and halts with an advisory that a SAR radar image is required.

7. Test Case: Professional Report Exports (GeoJSON & PDF)

Every completed analysis generates standard geospatial artifacts for GIS integration and auditing.

GeoJSON Export	Click **"Export GeoJSON"** ➔ Downloads valid FeatureCollection with reprojected EPSG:4326 WGS84 polygons ready for QGIS or ArcGIS.
PDF Audit Report	Click **"Export PDF"** ➔ Compiles full ReportLab report with audit trail, model versions, execution latency, and embedded high-resolution imagery.

SatQuery AI Test Suite Reference Document • Generated automatically from verified test matrices.
All demo assets are located locally under data/demo/ with valid geospatial CRS metadata.